



SCIENCE ORBIT

start is endless



1st term

$$01. p = \frac{a-t^2}{bx}$$

$$x = [L]$$

$$t = [T]$$

$$p = \frac{F}{A} = \frac{MLT^{-2}}{L^2} = [MT^{-1}T^{-2}]$$

$$[p \cdot bx] = [a] = [T^2]$$

$$[a] = T^2$$

$$\left[\frac{a}{b}\right] = \frac{T^2}{MT^{-1}T^{-2}}$$

$$= MT^2$$

$$[Anu = 2]$$

$$[p \cdot bx] = T^2$$

$$[b] = \frac{T^2}{MT^{-1}T^{-2} \times L} = MT^{-1}T^{-4}$$

$$2) E = kmv^2$$

$$\frac{\Delta E}{E} = \frac{\Delta m}{m} + 2 \frac{\Delta v}{v}$$

$$= 2 + 2 \times 3$$

$$= 2 + 6$$

$$= 8\%$$

3)

$$T = k \cdot p^a \cdot d^b \cdot F^c$$

$$p = F/A \quad E = J \text{ or } Nm$$

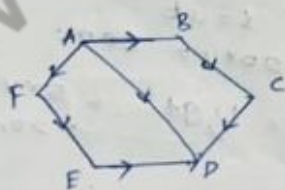
$$d = m/\omega$$

$$[T] = \left[\frac{MLT^{-2}}{L^2}\right]^a \left[\frac{M}{LT^{-1}}\right]^b [MLT^{-2} \cdot L]^c$$

$$a + b + c = 0$$

$$Ans = 2 \text{ only.}$$

4)



$$\vec{AF} + \vec{FE} + \vec{ED} = \vec{AD}$$

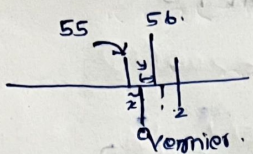
$$\vec{AB} + \vec{BC} + \vec{CD} = \vec{AD}$$

$$\therefore \vec{AD} + \vec{AD} + \vec{AD} = 3\vec{AD}$$

$$Ans = 5$$

05) $L.C = \frac{1}{10} = 0.1 \text{ mm}.$

$IND = IMD - LC$
 $= 0.9 \text{ mm}.$



$x + y = 1 \text{ mm}.$ (Length of vernier scale),

$x = 0.2 \text{ mm}.$

$\therefore y = 0.8 \text{ mm}.$

Ans = 5

Zero. 2nd Vernier 2nd division 2.25 mm
 Answer. Easy.

b) $5.5 \text{ mm} + 12 \times \frac{0.5}{50}.$

$5.5 \text{ mm} + 12 \times 0.01$

$5.5 + 0.12$

5.62 mm

$\rightarrow 0.562 \text{ cm}$

Ans = 2

a)

$S = ut$

$\Rightarrow 10 \mu\text{m/s} \times 100 \text{ s} = 10 \times 10^{-6} \times 100 = 1 \times 10^{-3} \text{ m}.$
 $= 1 \text{ mm}$

Ans = 1

08.



2s

So $V = 20 \text{ m/s}$

$\therefore V \cos 45^\circ = 20 \text{ m/s}$

$\rightarrow S = ut$

$20 \times 4 = 80 \text{ m}.$

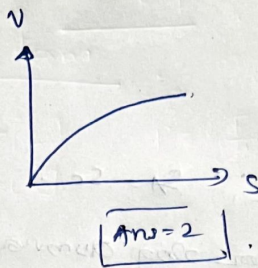
Ans = 1

$H = \frac{1}{2}gt^2 = 5 \times 4 = 20 \text{ m}.$

$\uparrow$ 20m 20m from origin

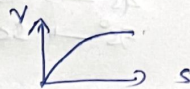
Ans = 1 from quick answer.

99) $V^2 = u^2 + 2as$
 $V^2 = 2as$
 $V^2 = \left(\frac{2g}{5}\right) \cdot s^2$
 $V = \sqrt{\frac{2g}{5}} \cdot s$

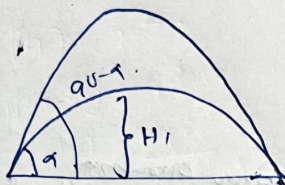


Ans = 2

$V^2 = 2as$
 $V = \sqrt{2as}$



10)



$\alpha + 90 - \alpha = 90$

so path is horizontal

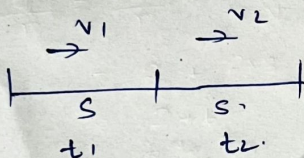
Maximum V is at the top

$(V \sin \alpha)^2 = 2gH_1 \rightarrow (1)$

$(V \sin(90 - \alpha))^2 = 2gH_2 \rightarrow (2)$

$\frac{H_1}{H_2} = \frac{\sin^2 \alpha}{\cos^2 \alpha}$ **Ans = 2**

11)



$S = v_1 t_1$

$t_1 = S/v_1$

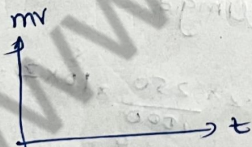
Dis. Ratio = $\frac{2000 \cdot 2000}{2000 \cdot 2000}$

$= \frac{2S}{t_1 + t_2}$

$= \frac{2S}{S/v_1 + S/v_2} = \frac{2v_1 v_2}{v_1 + v_2}$

Ans = 3

12)



Velocity = $\frac{mv}{t} = F = 2000 \text{ N}$

Momentum = $mv \cdot t = 2000 \text{ kg} \cdot \text{m/s} \cdot \text{s}$
 $= 2000 \text{ kg} \cdot \text{m}$

Ans = 4

13)

$$\begin{aligned} 15^\circ &\rightarrow S_A \\ 45^\circ &\rightarrow S_B \\ 75^\circ &\rightarrow S_C \end{aligned}$$

$$\therefore 15 + 75 = 90$$

$$\therefore S_A = S_C$$

$$45^\circ \text{ angle } \therefore S_A = S_C < S_B$$

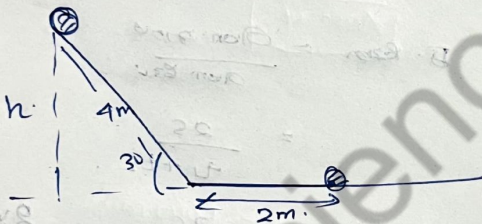
$$S_A = S_C \quad \text{but } B \text{ } 45^\circ \therefore H_A < H_C$$

$$H_A < H_B < H_C$$

$$\boxed{\text{Ans} = 5}$$

Longer drawing without calculation.

14)



$$\sin 30 = \frac{h}{4}$$

$$\frac{1}{2} = \frac{h}{4}$$

$$h = 2m$$

$$mgh - f \times d$$

$$\Rightarrow \frac{250}{1000} \times 10 \times 2 - \mu mg d$$

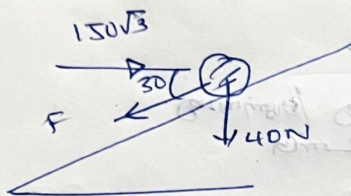
$$\frac{250}{1000} \times 10 \times 2 - 0.2 \times \frac{250}{1000} \times 10 \times 2$$

$$5 - 1$$

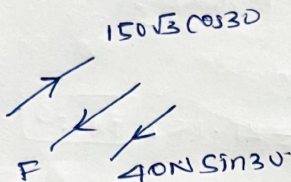
$$4J$$

$$\boxed{\text{Ans} = 4}$$

15)



20m/s (F) direction
20m/s 20m/s
20m/s.



$$225 - 20 = 205 (F).$$

$$F = 205N$$

$$\boxed{Ans = 3}$$

16)

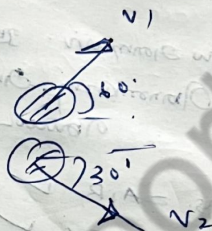
→ 24



2m
A



2m
B



2m/s 24m/s = 2m/s 24m/s

$$0 = 2m v_1 \cos 30^\circ - 2m v_2 \cos 60^\circ$$

$$v_1 \cos 30^\circ = v_2 \cos 60^\circ$$

$$\frac{v_1 \times \frac{\sqrt{3}}{2}}{2} = \frac{v_2 \times \frac{1}{2}}{2}$$

$$\frac{v_1}{v_2} = \frac{1}{\sqrt{3}}$$

$$v_2 = \frac{v_1}{\sqrt{3}}$$

$$v_2 = \sqrt{3} v_1$$

$$\rightarrow 2m u = 2m (v_1 \cos 60^\circ + v_2 \cos 30^\circ)$$

$$u = \left(v_1 \times \frac{1}{2} + \sqrt{3} \times \frac{\sqrt{3} v_1}{2} \right)$$

$$u = v_1/2 + 3/2 v_1$$

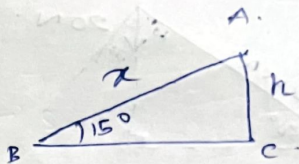
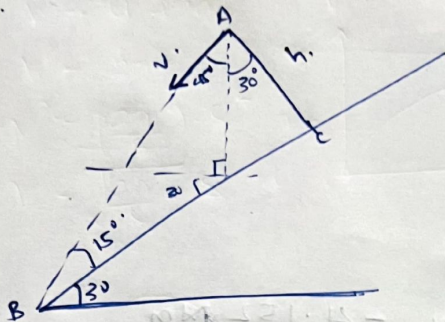
$$u = 2v_1$$

$$v_1 = u/2$$

$$\boxed{Ans = 1}$$

AM. Hruik.
BSc. Hons. Eng.

— ALC. Sammanthuni —



$$\sin(15) = \frac{h}{x}$$

$$x = \frac{h}{\sin 15}$$

$$S = ut$$

$$\frac{h}{\sin 15} = v \cdot t$$

$$t = \frac{h}{v \sin 15}$$

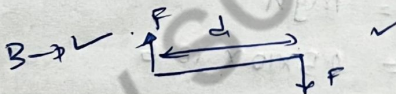
$$\boxed{\text{Ans} = 2}$$

$AB = x$. Longest time taken by the object to reach B.

12)

A → ✓

Object will move towards B with 2 m/s.

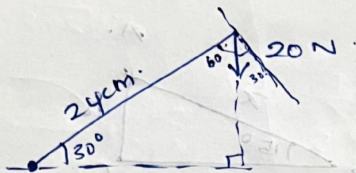


C → ? (Translation Issue)

but By the way, the object will move towards B with 2 m/s. This is the only possible answer. Correct ✓

Ans → A, B ✓

Ans → A, B, C ✓



$$Z = F \times 10830 \times 74$$

$$= 20 \times \frac{\sqrt{3}}{2} \times 24 = 4.152 \text{ Nm}$$

$$= 4.2 \text{ Nm} \quad \boxed{\text{Ans} = 2}$$

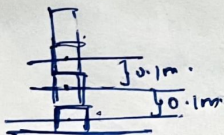
20) - Total work = $m \times g \times (h_1 + h_2 + \dots + h_n)$

Հանդիսի 2 մ շահարկող ուղղորդված շարժումն առաջացրած ընդհանուր աշխատանքի հաշվարկը:

$$\text{total height} = (0.1 + 0.2 + 0.3 + \dots + 1.4) \text{ m.}$$

$$= 0.1 (1 + 2 + \dots + 14)$$

$$= 0.1 \times \frac{14 \times 15}{2} = 10.5 \text{ m.}$$



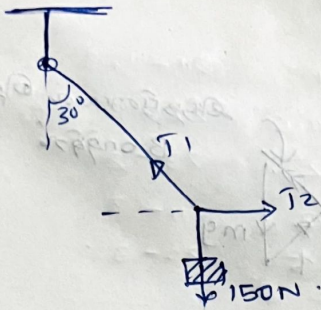
$$\therefore \text{total work} = mgh$$

$$= 0.5 \times 10 \times 10.5$$

$$= 52.5 \text{ J}$$

$$\boxed{\text{Ans} = 5}$$

21)



Am. Hazik.
Bse. Hons. Eng.

← Bofgon

$$T_1 \cos 60 = T_2$$

$$T_1 \times \frac{1}{2} = T_2$$

$$\frac{T_1}{T_2} = 2$$

$$\therefore \frac{T_2}{T_1} = \frac{1}{2}$$

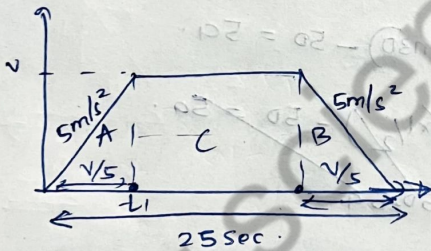
Ans = 3



1, 3 same

Ans = 1

22)



$$A+B = 2 \times \frac{1}{2} \times t_1 \times v$$

$$= 5t_1^2$$

$$72 \text{ km/h} \times \frac{5}{18} = 20 \text{ m/s}$$

$$C = v \times (25 - \frac{2v}{5})$$

$$= 5t_1 (125 - 10t_1)$$

$$= t_1 (125 - 10t_1)$$

$$20 \text{ m/s} = \frac{0 + 20 \text{ m/s}}{0.5 \text{ s}}$$

$$\frac{v}{t_1} = 5$$

$$t_1 = \frac{v}{5}$$

$$v = 5t_1$$

$$20 \text{ m/s} = \frac{(5t_1^2 + 125t_1 - 10t_1^2)}{25}$$

$$500 = -5t_1^2 + 125t_1$$

$$t_1^2 - 25t_1 + 100 = 0$$

$$(t_1 - 20)(t_1 - 5) = 0$$

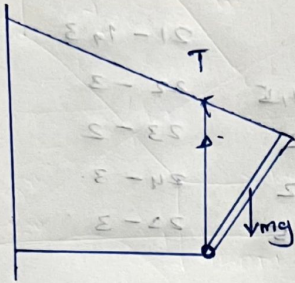
$$t_1 = 20, t_1 = 5$$

$$25 - 10$$

$$= 15 \text{ s}$$

Ans = 3

23)



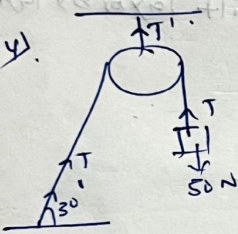
විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

Ans = 2

Am: Hazile
BSC: Hons Eng
ALC: STR

24)



විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

$$T + T \sin 30 = 120 \text{ N} = T$$

$$T + T/2 = 120$$

$$3T/2 = 120$$

$$T = 80 \text{ N}$$

Ans = 3

විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

$$80 - 50 = 5 \times 9$$

$$u = \frac{30}{5} = 6 \text{ m/s}^2$$

25)

A - ප්‍රදාන භෞතික විද්‍යා
B බ්ලොක්

විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

B - විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

C - විශිෂ්ට භෞතික විද්‍යා
B බ්ලොක්

$$F = \frac{20 \text{ mV}}{t}$$

Ans = A/C 3

My Answers

Q:

$$1-2$$

$$2-4$$

$$3-2$$

$$4-5$$

$$5-5$$

$$6-2$$

$$7-1$$

$$8-1$$

$$9-2$$

$$10-2$$

$$11-3$$

$$12-4$$

$$13-5$$

$$14-1$$

$$15-3$$

$$16-1$$

$$17-4,5$$

$$18-2$$

$$19-2$$

$$20-5$$

$$21-1,3$$

$$22-3$$

$$23-2$$

$$24-3$$

$$25-3$$

By. Am. Harik.
BSc. Hons. Buz.

-0759330924-

Note: please double check.
my arithmetic mistake. If any has.

→ Physics isn't learned in a day. It's built layer by layer.



$$E = 2\pi f$$

$$P \times 2 = 0.2 - 0.3$$

$$\text{and } = \frac{0.3}{2} = 0.15$$



3

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